NBM
Nordic Balancing Model

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Nordic FRR balancing capacity market

Nordic MMS

BSP implementation guide

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Revision history

Ver.	Date	Description
1.0	20.08. 2025	First released version covering FRR • New interface versions – definition of the ECP message type • Publishing reserve requirements 4.6 ○ process.processType ○ subject_MarketParticipant – removed ○ explanatory text for Bid_TimeSeries ○ quantity_Measurement_Unit.name - implementation change required when using the new IEC CIM scheme • Bid submission: ○ Delivery scenarios ○ connecting_Domain.mRID – improved description ○ status – improved description ○ quantity_Measurement_Unit.name - implementation change required when using the new IEC CIM scheme ○ minimum_ConstraintDuration.duration – improved description • Publishing bid acceptance results from Nordic MMS 4.7 - implementation changes required when using the new IEC CIM scheme: ○ Delivery scenarios ○ explanatory text for Bid_TimeSeries ○ bid_Original_MarketDocument.bid_**Bid**TimeSeries.mRID ○ quantity_Measurement_Unit.name ○ price_Measurement_Unit.name ○ code • In case that there are no reserve requirements defined for an auction, the empty CIM xml messages with just a header is sent. ○ Correction of the description • Bid updates to Nordic MMS is also allowed via ReserveBid_MarketDocument in version 7.4 (see 2.4 and 4.7)
1.1	14.11. 2025	New description in the chapter 4.8: publication of the quantity rejected due to bottleneck. New explanatory text: Bid_TimeSeries, Reason.code, Point.quantity New description of the domain.mRID element No new implementation is needed.
1.2	04.12. 2025	New description of the exclusive groups in the chapter 0. No new implementation is needed.

Ver.	Date	Description
1.3	19.03.2026	Changes to how indivisible bids are defined (chapters 3.3.5, 3.3.8, 4.7). Removal of the descriptions of the features which are currently not used on aFRR/mFRR markets. Formal cleaning.
1.4	08.10.2026	Specification of the dates from which changes to indivisible offers and exclusive linking of bids will be active.

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1 Introduction

1.1 Background

Nordic MMS has been developed to facilitate common Nordic capacity markets, with purpose to ensure the availability of reserves to ensure operational security. Nordic MMS allows TSOs to a joint procurement of balancing capacity in the Nordic market area before the day ahead market runs. Nordic MMS supports aFRR, mFRR and FCR reserve capacity markets, however the FCR reserve capacity markets are not the subject of this document.

Automatic frequency restoration reserve (aFRR), manual frequency restoration reserve (mFRR) and frequency containment reserve (FCR) are types of balancing services that TSOs use for balancing of the Nordic synchronous area. In the Nordic aFRR and mFRR capacity market, aFRR/mFRR balancing capacity is procured before the day-ahead market, considering geographical distribution and network constraints.

Nordic MMS allows the consideration of geographical distribution of reserve requirements, bids and network constraints in the market clearing. For FRR, the clearing uses the CZC reservation methodology.

The selection of bids (and reservation of cross-zonal capacity in the case of FRR) are done in an optimisation function maximizing socioeconomic benefit by weighing savings in reserve procurement through exchange of balancing capacity against the socio-economic cost incurred by reserving CZC from the day ahead electricity market.

1.2 Scope

This document covers the technical implementation details on the BSP side, when integrating with Nordic MMS by describing the processes required to interact with the FRR capacity markets. Both functional and technical aspects are covered. The intended audience of this document are the participating BSPs. In control areas where the BSP role has not been separated from the BRP role, the target audience for this document are the BRPs providing balancing services to the respective TSOs. Thus, the acronym BSP can be seen as synonymous to BRP in these cases.

The main processes described are:

- reserve requirement publication process,
- bid submission process,
- market result publishing process.

1.3 Terms and definitions

Acronym	Term	Definition
aFRR	Automatic frequency restoration reserves	The FRR that can be activated by an automatic control device designed to reduce the FRCE/ACE to zero
aFRR CM	aFRR Capacity Market	The aFRR capacity market's role in the balancing / ancillary service process is to procure obligations on behalf of the TSOs from the BSPs, in which the BSPs commit to offering the allocated amount of balancing energy bids in the aFRR activation market.

Acronym	Term	Definition
mFRR	Manual frequency restoration reserves	Manual frequency restoration reserves: active power reserves that may be manually activated, available to restore system frequency to the nominal frequency and, for a synchronous area consisting of more than one LFC area, to restore power balance to the scheduled value
mFRR CM	mFRR Capacity Market	The mFRR capacity market's role in the balancing / ancillary service process is to procure obligations on behalf of the TSOs from the BSPs, in which the BSPs commit to offering the allocated amount of balancing energy bids in the mFRR activation market.
BRP	Balance Responsible Party	A market participant or its chosen representative responsible for its imbalances
BSP	Balancing Services Provider	A market participant with reserve-providing units or reserve-providing groups able to provide balancing services to TSOs
CCC	Coordinated Capacity Calculation	The Nordic RSC's is responsible to calculate cross-border transmission capacities to maximise the transmission capacity offered to the day-ahead market.
CIM	IEC Common Information Model	
CZC	Cross Zonal Capacity	The cross-zonal transmission capacity between two bidding zones
ECP	Energy Communication Platform	Reference implementation of MADES standard
MADES	Market Data Exchange Standard	Communication IEC standard designed by ENTSO-E
MOL	Merit Order List	
MTU	Market Time Unit	The period for which the market price is established or the shortest possible common time period for the two bidding zones if their market time units are different.
TSO	Transmission System Operator	A party that is responsible for a stable power system operation (including the organisation of physical balance) through a transmission grid in a geographical area. In the Nordic synchronous area, there are four TSOs: Svenska kraftnät, Fingrid, Energinet.dk and Statnett.
	connecting TSO	the TSO that operates the scheduling area in which balancing service providers and balance responsible parties shall be compliant with the terms and conditions related to balancing;

1.4 References

Ref [1] The Nordic aFRR Capacity Market rules

Ref [2] The Nordic mFRR Capacity Market rules

Ref [3] Nordic Balancing Model – Publications (Implementation guides for BSP, etc.)

Ref [4] Energy Market Communications – Acknowledgement Business Process and Contextual Model for CIM European Market (IEC 62325-451-1)

Ref [5] ENTSO-E Reserve Bid document – UML model and schema v 7.1

Ref [6] ENTSO-E Reserve Bid document – UML model and schema v 7.4

Ref [7] ENTSO-E Balancing Document – UML model and schema v 4.5

Ref [8] ENTSO-E Reserve Allocation Result document – UML model and schema v 6.4

Ref [9] ENTSO-E Acknowledgement document

2 Business context

2.1 aFRR market timeline

The aFRR capacity market involves several operational phases. The diagram below shows the timeline for the aFRR capacity market.

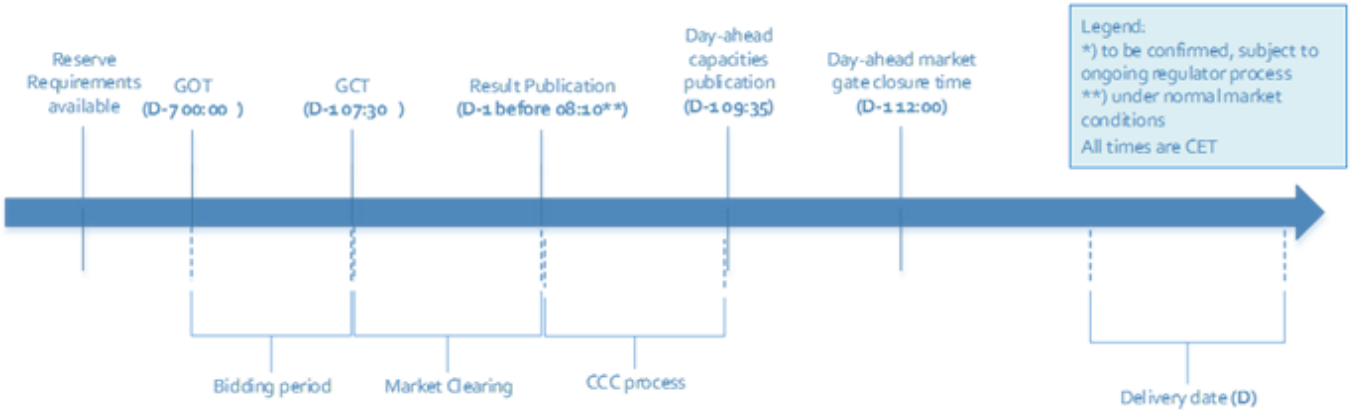


Figure 1: aFRR market timeline

2.2 mFRR market timeline

The mFRR capacity market involves several operational phases. The diagram below shows the timeline for the mFRR capacity market.

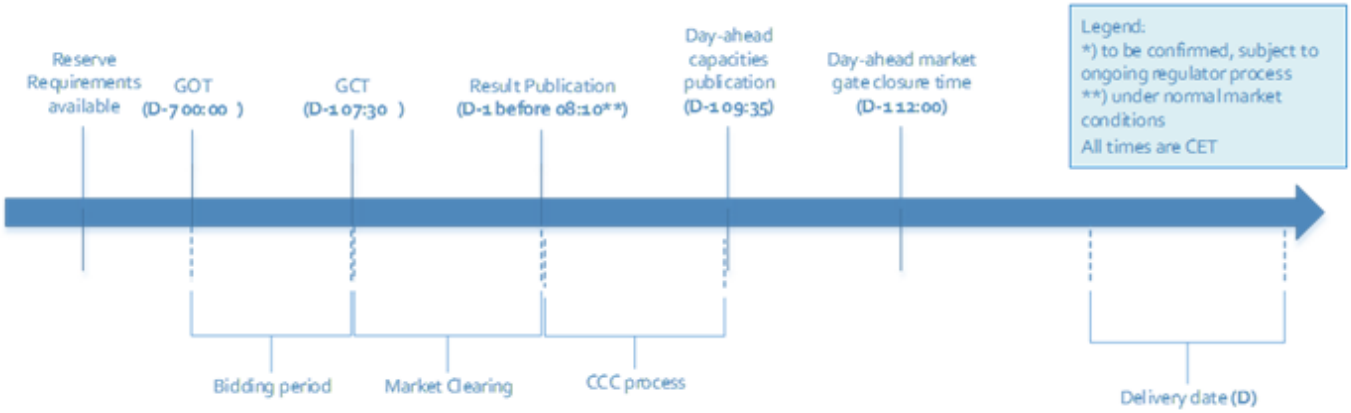


Figure 2: mFRR market timeline

2.3 Timeline description

- Long-term reserve requirements are published in regular intervals on the TSOs websites and visible in Nordic MMS.
- At the FRR capacity markets Gate Opening Time (GOT), Nordic MMS publishes the current reserve requirements of the delivery day of the opened auction to TSOs and BSPs.
- FRR markets gate opening time: BSPs can submit their bids to all capacity markets from D – 7 00:00 CET.

- Subsequently, when reserve requirements are updated, Nordic MMS republishes them to BSPs and TSOs up until the gate closure time (GCT).
- Based on whether reserve requirements are updated after gate opening, the TSOs publishes reserve requirements through the described reserve requirement publishing process.
- Between the capacity market Gate Opening Time (GOT) and the capacity market Gate Closure Time (GCT), the BSPs can submit and update their bids (Bidding period).
- FRR capacity markets gate closing time: BSPs can submit their bids to the aFRR / mFRR market until D – 1, 07:30 CET.
- After gate closure, the market clearing runs. The bid selection optimisation is executed, and the market results are published to BSPs and TSOs.
    - mFRR result publication: no later than D – 1, 8:10 CET
    - aFRR result publication: no later than D – 1, 9:20 CET
- The TSOs have the possibility to extend the bidding period or reopen the bidding gate.
- Day-ahead aFRR/mFRR capacities publication:
    - mFRR: At 9:35 CET, available capacities for common day ahead market coupling are published by the RSC
    - aFRR: At 10:00 CET, available capacities on interconnectors and in the grid are published

2.4 System context

This document provides detailed information about the message exchanges between Nordic MMS and the BSPs. The other exchanges are shown for contextual information purposes only and are outside the scope of this document.

The table below provides an overview of the flows between Nordic MMS and the BSPs. The table shows that IEC CIM is used as data exchange format and ECP/ EDX is used as communication platform. Please refer to chapter 5 for more details related to ECP/EDX.

Sender	Receiver	Data	Document	ECP message type
Nordic MMS	BSP	Publish reserve requirements	iec62325-451-7-reservebiddocument.xsd – version 7.4	MO-MCC-RESERVE-REQUIREMENTS-V7-4
BSP	Nordic MMS	Bids	iec62325-451-7-reservebiddocument.xsd – version 7.1 iec62325-451-7-reservebiddocument.xsd – version 7.4	MO-MCC-BIDS
Nordic MMS	BSP	Acknowledgement	iec62325-451-1-acknowledgementdocument.xsd – version 8.0	"original message code"-ACK

Sender	Receiver	Data	Document	ECP message type
BSP	Nordic MMS	Acknowledgement	iec62325-451-1-acknowledgementdocument.xsd – version 7.0 iec62325-451-1-acknowledgementdocument.xsd – version 8.0 („preferred“) iec62325-451-1-acknowledgementdocument.xsd – version 8.1	"original message code"-ACK
Nordic MMS	BSP	Accepted bids	iec62325-451-7-reserveallocationresult.xsd – version 6.4	MO-MCC-ACCEPTED-BIDS-BSP-V6-4
Nordic MMS	TSO	Market result	iec62325-451-6-balancing.xsd – version 4.5	MO-MCC-MARKET-RESULTS-V4-5

Table 1 Flow between Nordic MMS and BSP

2.5 General rules

The capacity markets business process flows assume certain basic rules. These rules are described below.

2.5.1 Date and time

Date and time are expressed using the standard XML format for date and time: `YYYY-MM-DDTHH:MM:ssZ`, formatted using the universal time standard UTC by adding a *'Z'* behind the time - like this: 2022-06-14T22:00:00Z

2.5.2 Document coverage

The beginning and ending date and time of the period covered by the document shall cover just one CET/CEST day.

2.5.3 Daylight saving time

- In winter, the period is from 23:00 UTC to 23:00 UTC
- In summer, the period is from 22:00 UTC to 22:00 UTC
- On the date of the change from winter to summer time, the period is from 23:00 UTC to 22:00 UTC. This change occurs on the last Sunday in March at 01:00 UTC
- On the date of the change from summer to winter time, the period is from 22:00 UTC to 23:00 UTC. This change occurs on the last Sunday in October at 01:00 UTC

2.5.4 Document identification and revision number

The document identification must be unique over time for the sender in question. Furthermore, the document identification itself should not have any significant meaning. The revision number is not used and shall always be equal to '1'.

2.5.5 Update/cancel principles

In general, a new received document will always completely replace a previously received document. Update of any time series is done by sending a new document honouring these rules:

- A new document mRID (document identification)
- The same revision number (always equal to '1')
- A newer created date-time
- The same period/day, domain, auction.mRID and subject market participant (for bids) as for the data being updated

I.e. to cancel one or more time series, a new document is submitted honouring the above rules, omitting the bids that should be cancelled.

2.5.6 Cancel all bids

In order to cancel all bids by a given sender, day, and control area, a new document should be submitted with a dummy Bid_TimeSeries using element "status" and value A09 = cancelled. The time series must include the element auction.mRID, with the valid market code.

- The document period is specified as usual using the reserveBid_Period.timeInterval element.
- The document's scope (control area/bidding zone) is specified using the domain.mRID element.
- The dummy timeseries must include the status element and a reference to the market using element auction.mRID.
- The content of the other mandatory timeseries elements is ignored.

Working example:

- targetNamespace="urn:iec62325.351:tc57wg16:451-7:**reservebiddocument:7:1**"

```
<ReserveBid_MarketDocument>
…
  <Bid_TimeSeries>
      <mRID>8c520815945a43e88e1fb72e9ad05507</mRID>
      <auction.mRID>MFRR_CAPACITY_MARKET</auction.mRID>
      <businessType>B74</businessType>
      <acquiring_Domain.mRID codingScheme="A01">10Y1001A1001A91G</acquiring_Domain.mRID>
      <connecting_Domain.mRID codingScheme="A01">10YNO-3--------J</connecting_Domain.mRID>
      <quantity_Measure_Unit.name>MAW</quantity_Measure_Unit.name>
      <currency_Unit.name>EUR</currency_Unit.name>
      <price_Measure_Unit.name>MAW</price_Measure_Unit.name>
      <divisible>A01</divisible>
      <blockBid>A02</blockBid>
      <status>
         <value>A09</value>
      </status>
      <flowDirection.direction>A01</flowDirection.direction>
      <marketAgreement.type>A01</marketAgreement.type>
      <Period>
         <timeInterval>
            <start>2000-01-01T00:00Z</start>
            <end>2000-01-01T00:00Z</end>
         </timeInterval>
         <resolution>PT60M</resolution>
         <Point>
            <position>1</position>
            <quantity.quantity>0</quantity.quantity>
            <price.amount>0</price.amount>
         </Point>
      </Period>
  </Bid_TimeSeries>
</ReserveBid_MarketDocument>
```

- targetNamespace="urn:iec62325.351:tc57wg16:451-7:**reservebiddocument:7:4**"

```
<ReserveBid_MarketDocument>
…
  <Bid_TimeSeries>
      <mRID>8c520815945a43e88e1fb72e9ad05507</mRID>
      <auction.mRID>MFRR_CAPACITY_MARKET</auction.mRID>
      <businessType>B74</businessType>
      <acquiring_Domain.mRID codingScheme="A01">10Y1001A1001A91G</acquiring_Domain.mRID>
      <connecting_Domain.mRID codingScheme="A01">10YNO-3--------J</connecting_Domain.mRID>
      <quantity_Measurement_Unit.name>MAW</quantity_Measure_Unit.name>
      <currency_Unit.name>EUR</currency_Unit.name>
      <price_Measurement_Unit.name>MAW</price_Measure_Unit.name>
      <divisible>A01</divisible>
      <blockBid>A02</blockBid>
      <status>
         <value>A09</value>
      </status>
      <flowDirection.direction>A01</flowDirection.direction>
      <marketAgreement.type>A01</marketAgreement.type>
      <Period>
         <timeInterval>
            <start>2000-01-01T00:00Z</start>
            <end>2000-01-01T00:00Z</end>
         </timeInterval>
         <resolution>PT60M</resolution>
         <Point>
            <position>1</position>
            <quantity.quantity>0</quantity.quantity>
            <price.amount>0</price.amount>
         </Point>
      </Period>
  </Bid_TimeSeries>
</ReserveBid_MarketDocument>
```

2.5.7 Acknowledgement

For the *Bid submission* process, an acknowledgement document, as defined in Ref [9] is generated as response, either completely accepting the received document or rejecting it completely. Partial accept of documents is not used. The codes and values used in this document are provided in chapter 4.10.

The acknowledgement will always contain a document level `Reason`, with either code `A01` (Message fully accepted) or `A02` (Message fully rejected).

The acknowledgement may also contain further document level `Reason`(s), explaining document level errors.

The acknowledgement may also contain one or more `Rejected_Timeseries`, with one or more timeseries level `Reason`(s).

The acknowledgement may also contain one or more `InError_Period` within the `Rejected_Timeseries`, with one or more `Reason`(s) for a given time interval.

See example code snippet below:

```
<Acknowledgement_MarketDocument>
…

  <Rejected_TimeSeries>
      <mRID>4CDF6AAA-4C0D-98DB-94CDE58FB4B5</mRID>
      <InError_Period>
          <timeInterval>
              <start>2021-01-07T00:00Z</start>
              <end>2021-01-07T01:00Z</end>
          </timeInterval>
          <Reason>
              <code>A59</code>
              <text>All quantities of block bid must be equal.</text>
          </Reason>
      </InError_Period>
      <Reason>
        <code>A22</code>
        <text>Invalid BSP</text>
```

```
      </Reason>
  </Rejected_TimeSeries>
  <Reason>
    <code>A02</code>
    <text>Document fully rejected.</text>
  </Reason>
  <Reason>
    <code>A51</code>
    <text>The attribute createdDateTime cannot be in the future.</text>
  </Reason>
<Acknowledgement_MarketDocument>
```

2.5.8 Time series period

Within a time series period, the position must always begin with '1' and be incremented by '1' for each subsequent position, forming a series of contiguous numbers covering the complete range of the period. Furthermore, *gaps* in the time series are allowed. In this case, the time series will comprise of several periods to indicate the gaps.

See code snippet below:

```
<Bid_TimeSeries>
...
    <Period>
        <timeInterval>
            <start>2018-09-17T03:00Z</start>
            <end>2018-09-17T05:00Z</end>
        </timeInterval>
        <resolution>PT60M</resolution>
        <Point>
            <position>1</position>
            <quantity.quantity>100</quantity.quantity>
            <minimum_Quantity.quantity>100</minimum_Quantity.quantity>
            <price.amount>25.20</price.amount>
        </Point>
        <Point>
            <position>2</position>
            <quantity.quantity>100</quantity.quantity>
            <minimum_Quantity.quantity>100</minimum_Quantity.quantity>
            <price.amount>25.20</price.amount>
        </Point>
    </Period>
    <Period>
        <timeInterval>
            <start>2018-09-17T15:00Z</start>
            <end>2018-09-17T17:00Z</end>
        </timeInterval>
        <resolution>PT60M</resolution>
        <Point>
            <position>1</position>
            <quantity.quantity>100</quantity.quantity>
            <minimum_Quantity.quantity>100</minimum_Quantity.quantity>
            <price.amount>25.20</price.amount>
        </Point>
        <Point>
            <position>2</position>
            <quantity.quantity>100</quantity.quantity>
            <minimum_Quantity.quantity>100</minimum_Quantity.quantity>
            <price.amount>25.20</price.amount>
        </Point>
    </Period>
</Bid_TimeSeries>
```

3 Business process

This chapter provides information related to the business rules that apply.

3.1 Sequence diagram

The sequence diagram for the whole capacity market is depicted below. Note that the last two shown exchanges are provided for information only and are outside the scope of this document. Please refer to local user guides for further details.

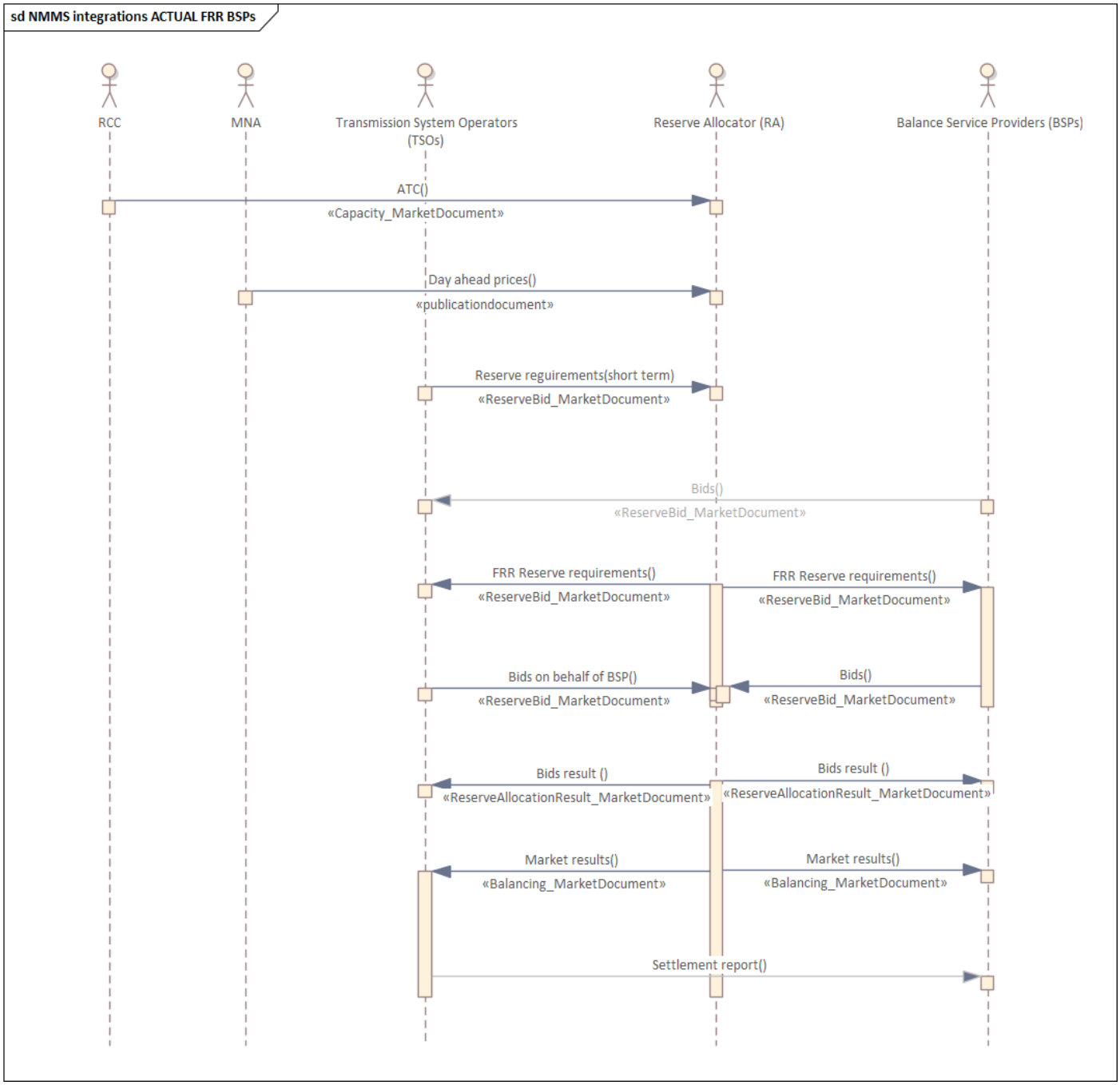


Figure 1 Sequence diagram for the capacity markets

3.2 Reserve requirements

Publication of the reserve requirements is maintained by the MO (market operator) as long-term requirements in Nordic MMS directly.

Nordic MMS publishes reserve requirements as CIM xml messages at gate opening to inform the BSPs and TSOs about the reserve requirements known at the moment when the auction is being opened. TSOs can update daily reserve requirements before auction clearing, and Nordic MMS will then publish the updated reserve requirements to all TSOs and BSPs.

In case that there are no reserve requirements defined for an auction, the empty CIM xml messages with just a header is sent.

In Nordic MMS geographic areas that can carry reserve requirements can be bidding zones and / or macro areas in case of the aFRR and mFRR markets.

Macro areas are sets of one or more bidding zones. The minimum and maximum regulation may be defined as a part of the reserve requirements definition on a macro area and if specified, it works as a localization constraint, i.e. it sets a minimum or maximum limit on how much of the bid volume can be selected locally in the macro area's set of bidding zones. This can result in forcing or limiting exchange between bidding zones.

In CIM messages, minimum and maximum regulations that function as localization constraint in addition to the need, will appear in additional time series with their respective business types.

This process is used for the distribution of the reserve requirements (demand, minimum regulation and maximum regulation) of bidding zones and macro areas towards BSPs. The provided data is considered public information and all BSPs receive the same document.

The published reserve requirements are daily data in hourly resolution. They apply to a single auction and delivery day only.

Reserve requirements are published shortly after gate opening. They are also sent out as result of being updated after they have been initially published, which allows TSOs to make and inform about changes even after the bidding gate has been opened.

3.3 Bid submission

During the bidding phase, the participating BSPs provide all information related to bids for the FRR capacity markets. The *ReserveBid_MarketDocument* is used to provide this information. Alternatively, bids can be submitted via the Nordic MMS web user interface. The same business rules apply for both alternatives.

In order to be able to submit bids to Nordic MMS, the BSP needs to have a valid portfolio in the market and bidding zone to bid in. A portfolio represents that the BSP has prequalified resources to provide the required type of balancing services in a certain bidding zone and an agreement with the responsible TSO to participate in the market.

For a general description of the ReserveBid_MarketDocument schema, please refer to Ref [5]. The codes and values to be used in this document are provided in chapter 4.7.

The following business rules apply to the bid submission process. It is up to the BSPs' responsible TSO to decide whether some the following market features are enabled or not as well as setting threshold values.

3.3.1 General rules

- A Reserve Bid Document contains a set of bids (a bid is represented by a time series)
- A Reserve Bid Document shall be for only one *subject party*
- The ECP endpoint used to send the data must be associated with the subject party either by:
  - The subject party being the same as the sender and being directly associated with endpoint used
  - The subject party having an active agent-principal (BSP) relationship with the agent identified in the sender market participant (logical sender) and by the sender ECP endpoint (physical sender)
- The delivery period for a bid must belong to the same tender period for all bids in a Reserve Bid Document
- Bids are given per bidding zone, which means as portfolio-bids, unless the TSO requires the BSPs to provide further location information on the bid using registered resource identifiers:
  - Statnett requires locational information both for aFRR and mFRR.
  - Energinet and Fingrid require locational information for mFRR.
- One Reserve Bid Document can be sent per control area, which includes all bids for all bidding zones the BSP has a portfolio in. It is also possible to send Reserve Bid Document per bidding zone. The document scope (control area or bidding zone) defines the set of bids replaced by the new document sent in. The number of bids submitted may not exceed the maximum number of bids set per portfolio.

3.3.2 Bid quantity

- Bid quantity may vary for all hours in the time series
- Bid quantity (and minimum bid quantity) must be an integer number between 0 and 999.
- Bid quantity may not exceed the [Maximum bid quality] per regulation direction and bidding zone, given by the BSP portfolio.

3.3.3 Bid price

- Bid price must be present and have the same value for all positions in a time series
- Bid price must be between 0 and 9,999 €/MW/h, with maximum two decimal places.

3.3.4 Block bids

- All quantities of a block bid must be equal and represent a continuous interval of market time units - no gap is allowed in the sequence of MTUs.
- All offered quantities of the block bid must be accepted or rejected.
- A block bid cannot be part of an exclusive group.

3.3.5 Bid divisibility

The divisible bid is the basic bid type. The bid can be accepted or rejected fully or partially, i.e. the accepted quantity is between offered quantity and zero, however accepted quantity different from zero cannot be less than minimum quantity of the bid. Minimum quantity of a divisible bid must not exceed 50 MW.

An indivisible bid must be accepted or rejected fully, i.e. the accepted quantity is equal to the offered quantity or zero. Minimum quantity of the bid must not be set for indivisible bids. Offered quantity of an indivisible bid must not exceed 50 MW.

Bids marked as indivisible may be submitted only for the delivery days before 27.10.2026. Since that day, indivisibility in a bid can only be induced by the BSP setting the minimum quantity appropriate (but not higher than 50 MW). Exclusive linked bids

Two or more bids can be linked into single exclusive group. For each MTU, only one bid which is part of the exclusive group can be accepted, all other bids must be rejected in that MTU.

Nordic MMS supports exclusive linking of bids within the same market and between aFRR and mFRR market. Markets are cleared in a specific clearing order: aFRR CM before mFRR CM. The rules for the cross-market exclusive linking change as of the delivery date 15.10.2026.

Rules for cross-market exclusive linking valid till the delivery date 15.10.2026:

> If an aFRR bid which is part of an exclusive group across markets is selected, all mFRR bids in the same exclusive group will not be offered to the bid selection of the mFRR market and can thus not be selected there in the same market time unit.

Rules for cross-market exclusive linking valid since the delivery date 15.10.2026:

> If an aFRR bid which is part of an exclusive group across markets is selected, all divisible mFRR bids in the same exclusive group will be offered to the bid selection of mFRR market with the offered quantity decreased by the accepted quantity of the aFRR bid. If the quantity after decreasing would be less than minimum quantity, the corresponding bid becomes unavailable.

Please note that the common order of the market clearings (first aFRR, second mFRR) may exceptionally be reversed. The above-described rules apply then analogically from mFRR to aFRR.

Rules for creation of the exclusive group:

- Bids in the same exclusive group must belong to the same bidding zone (*connecting_Domain.mRID*)
- Bids in the same exclusive group must be of the same direction
- An exclusive group must contain at least two bids
- A block bid cannot be part of an exclusive group.

Evaluation of the exclusive groups within the same market:

- Single hour bids:
  - The single hour bid is evaluated separately for each MTU => multiple single hour bids can be accepted but accepted MTU will be always different

	Exclusive group	Block bid	1:00	2:00	3:00	4:00	
BID1	2	✖	1	1	1		Single hour bids only
BID2		✖	1		1		

	Exclusive group	Block bid	1:00	2:00	3:00	4:00	
BID1	3	✖	1	1	1	1	Single hour bids only
BID2		✖	1		1	1	

Legend:

- Green colour: the bid is accepted in the given hour
- Red colour: the bid cannot be accepted because of acceptation of the other bid in the given hour
- No colour: no acceptation = no restriction

3.3.6 Technical linking and quality limitations (technically linked group)

The feature allows the following bid quality constraints to be set:

- Resting time
- Maximum activation duration
- Quality constraints can be set for the bids for the mFRR service.
- This feature is not allowed for the bids for the aFRR service.

Bid time series with common resting time and activation duration limitations need to be specified with a common technical linked group identification in the linkedBidsIdentification element. These bids are then chosen only in MTUs where the bids are available for activation.

The bid selection will not select such bids in market time units where they can be unavailable due to selections in other MTUs of the same or other time series with the same linkedBidsIdentification.

- Bids in the same technically linked group must be situated in the same bidding zone.
- Bids in the same technically linked group must have the same duration and resting time values.
- The technically linked group must contain at least two bids.
- Block bid cannot be part of technically linked group.

3.3.7 Registered resource ID

Nordic MMS allows specifying registered resource identifiers on the bids if the responsible TSO requires that information.

3.3.8 Allowed bid type combinations

Bids can be linked in time or mutually exclusive. They can be also technically linked. Before delivery day 25.11.2026, they can be also indivisible, but from delivery day 25.11.2026, the bid cannot be directly market as indivisible. The indivisibility may be, however, achieved by the setting a sufficiently high minimum quantity. The table below describes each of these bid properties, and every allowed combination of these properties.

Divisibility	Block (B)	Exclusive (E)	Technical linking and quality limitations	Description
Divisible	No	No	Technical linking available – mFRR only	Single bid – varying quantity – divisible, non-block bid has the same rules as described in case 0 with the difference that it can be accepted in the range between minimum and offered quantity. The divisibility is expressed by the presence of the minimum quantity. The minimum quantity can be also zero. The result can be different for every hour of the bid.

Divisibility	Block (B)	Exclusive (E)	Technical linking and quality limitations	Description
Indivisible (before 25.11.2026)	No	No	Technical linking available – mFRR only	Single bid – indivisible, non-block bid can span one or more hours (discontinuous interval is allowed). Each hour is cleared separately, i.e. the bid can be accepted or rejected separately hour by hour. The accepted quantity must either be equal to the offered quantity or zero. The result can be different in every hour of the bid.
Divisible	Yes	No	Technical linking not available	Divisible block (between minimum and offered quantity) bid spans multiple consecutive hours (discontinuous interval is not allowed) with the same quantities per hour. In all hours, the bid must be either accepted (between minimum and offered quantity) or rejected. I.e. it is not possible to accept the bid in one hour and reject it in another hour.
Indivisible (before 25.11.2026)	Yes	No	Technical linking not available	Indivisible block bid spans multiple consecutive hours (discontinuous interval is not allowed) with the same quantities per hour. In all hours, the bid must be either fully accepted or rejected.
Indivisible (before 25.11.2026)	No	Yes	Technical linking available – mFRR only	Exclusive bids in the same group (they have the same exclusive group id) are mutually exclusive for the same hour (block bids cannot be part of the exclusive bid group). The bid can be accepted in the given hour only if all other bids in the group are rejected. Bids can be exclusively linked across markets (aFRR/mFRR).
Divisible	No	Yes	Technical linking available – mFRR only	The same case as IE with the difference that the bid can be accepted also partially (between minimum and offered quantity). Please note that the exclusive group can contain both divisible and indivisible bids (before 25.11.2026). Bids can be exclusively linked across markets (aFRR/mFRR).

3.3.9 Invalid bid type combinations

Block (B)	Exclusive (E)	Description
Yes	Yes	Block bid cannot be part of the exclusive group.
Yes	Yes	Block bid cannot be part of the exclusive group.

3.4 Market result publishing

When the bid selection optimization process is completed, and the result from the auction is ready, the following information is published from the Nordic MMS system:

Document		Information	Timing
Accepted bids **ReserveAllocationResult_MarketDocument**		Only own bids Both procured and offered prices and quantities	When bid selection optimization process is completed and the auction results have been published.
Market result **Balancing_MarketDocument**	Volumes	Market volumes	
	Prices	Market prices	

Settlement of accepted bids uses the pay-as-cleared methodology, meaning that the market result includes clearing prices per bidding zone, market time unit and regulation direction with the price being included both in the balancing market document and the reserve allocation result market document.

3.4.1 Accepted bids

The *ReserveAllocationResult_MarketDocument* as defined in 4.7 is used to provide information about BSPs bids acceptation. Firstly, a status of the bids that submitted to the auction is provided after auction clearing.

One of the following reasons is used to indicate the status of the Auction contract (bids):

- bid is accepted (**`A73`**)
- bid is not accepted (**`B09`**)
- bid is partially accepted (**`A72`**)
- bid is rejected by TSO or due to the bottleneck (**`B16`**)

3.4.2 Market results – volumes and prices

The *Balancing_MarketDocument* as defined in 4.9 is used to provide information about the market prices and the total volume per hour and direction. The codes and values to be used in this document are provided in chapter 4.9. Nordic MMS distributes one *Balancing_MarketDocument* message per BSP and market result publication run. BSPs which did not enter bids into an auction do not by default receive market result messages. A BSP can however be flagged in Nordic MMS to receive the market results messages even if the BSP does not enter bids into an auction. To enable this, the BSP should contact their local TSO.

3.4.3 Republishing of market results

Nordic MMS can republish the market results multiple times due to different reasons:

- If the auction result is valid but cannot be processed by recipients due to inconsistencies in party identifiers. In this case, master data in the local MMS or in Nordic MMS requires to be updated before the market result is republished.
- Republishing can also be initiated when communication with some or all recipient endpoints has previously failed.

To support the update/cancel principle described in 2.5.5 and ensuring that the new market result replaces any earlier received market results, all documents that are sent will have new document mRID and updated createdDateTime.

3.4.4 Withdrawal of market results

Nordic MMS can withdraw the market result because of an invalid clearing result. As part of a withdrawal process, Nordic MMS will send out new market result messages with the following changes:

- Any previously accepted or partially accepted bids are changed to not accepted, which means the reason code is set to **B09 –** Bid not accepted
- All accepted volumes in the total market results are set to 0
- The price element is not included

Nordic MMS allows auction results to be published and withdrawn multiple times for one auction run.

3.4.5 Finish with non-purchase

Finish with non-purchase is a special market result, where:

- All reason codes in the accepted bids message are set to **B09 –** Bid not accepted
- All accepted volumes in the total market results are set to 0

There are three possible scenarios for such zero results to be distributed:

- There are no reserve requirements
- No auction result has been published within the final publication deadline
- The auction has been cancelled because of extraordinary market conditions or a permanent failure during market clearing

Non-purchase market results completely replace and invalidate earlier published results.

4 Data flow descriptions

This chapter provides the dependencies for the documents used to support the FRR and FCR capacity markets.

4.1 Classification

The dependency matrices describe the classification of the attributes. The following are classifications are used:

- M – The information is mandatory, i.e. the element is mandatory in the XSD schema
- R – The information is required, i.e. the element is not mandatory in the XSD schema but is required by the business process
- D – The information is dependent, i.e. the presence depends on certain condition(s)
- O – The information is optional.

4.2 Coding schemes

When communication with Nordic MMS, the coding scheme to identify senders and receivers can vary by TSO. The local TSO defines which coding scheme and party code to use for the market participants it is responsible for.

Some sender / receiver systems like Nordic MMS use fixed codes and coding scheme A01.

Code	Coding scheme	Description and reference
A01	EIC	Energy Identification Codes: https://www.entsoe.eu/data/energy-identification-codes-eic/
A10	GS1/ GLN	Global Location Number, provided by GS1: https://gepir.gs1.org/index.php/search-by-gln
NSE	Swedish national	
NFI	Finnish national	
NDK	Danish national	
NNO	Norwegian national	

4.3 Bidding zones

The FRR capacity market consists of the following bidding zones:

Name	TSO	mRID	Coding scheme
DK1	Energinet	10YDK-1--------W	EIC
DK2	Energinet	10YDK-2--------M	EIC
FI	Fingrid	10YFI-1--------U	EIC
NO1	Statnett	10YNO-1--------2	EIC
NO2	Statnett	10YNO-2--------T	EIC
NO3	Statnett	10YNO-3--------J	EIC
NO4	Statnett	10YNO-4--------9	EIC
NO5	Statnett	10Y1001A1001A48H	EIC
SE1	Svenska kraftnät	10Y1001A1001A44P	EIC
SE2	Svenska kraftnät	10Y1001A1001A45N	EIC
SE3	Svenska kraftnät	10Y1001A1001A46L	EIC
SE4	Svenska kraftnät	10Y1001A1001A47J	EIC

4.4 Control areas

- Denmark (DK): 10Y1001A1001A796
- Finland: 10YFI-1--------U
- Norway: 10YNO-0--------C
- Sweden: 10YSE-1--------K

4.5 Macro areas

- Norway: 10YNO-0--------C
- Sweden: 10YSE-1--------K
- Finland: 10YFI-1--------U

4.6 Publishing reserve requirements by Nordic MMS

The data flow is used for distribution of the reserve requirements (demand, minimum regulation and maximum regulation) of bidding zones and macro areas towards BSPs.

The publishing of reserve requirements uses the ReserveBid_MarketDocument in version 7.4.

For a general description of the ReserveBid_MarketDocument schema in version 7.4, please refer to Ref [6].

Document scope:

Each received document shall always contain data for one day and:

- all bidding zones assigned to the market for aFRR/mFRR markets, and
- all macro areas assigned to the market.

ReserveBid_MarketDocument		**451-7-reservebiddocument.xsd – version 7.4**
mRID	M	Unique identification of the document.
revisionNumber	M	Constant value of **1**
Type	M	**B21** = Reserve need document
process.processType	M	**A47** = Manual frequency restoration reserve **A51** = Automatic frequency restoration reserve
sender_MarketParticipant.mRID	M	10V1001C--000284 (Nordic MMS) **A01** = EIC coding scheme
sender_MarketParticipant.marketRole.type	M	**A34** = Reserve Allocator
receiver_MarketParticipant.mRID	M	Identification of the party receiving the document Identification is supported by several coding schemes. See Chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	M	**A46** = Balancing Service Provider (BSP)
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
reserveBid_Period.timeInterval	M	The period covered by the document (in ISO 8601 UTC format) Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ
domain.mRID	M	EIC identification of the market area: 10Y1001A1001A91G = Nordic Area **A01** – EIC coding scheme
BidTimeSeries		

Bidding zone requirements: Three time series may be present for each bidding zone and each direction: • **demand,** • **minimum regulation,** • **maximum regulation.** **Macro area requirements: Two time series may be present for each macro area and each direction:** • **minimum regulation,** • **maximum regulation.**		
mRID	M	Unique identification of the need timeseries.
auction.mRID	M	Constant value: MFRR_CAPACITY_MARKET AFRR_CAPACITY_MARKET
businessType	M	For demand: **B75** = The demand that should be covered in the specified amount. Relevant for a bidding zones and macro areas in the acquiring domain. For minimum regulation: **A60** = A localization constraint for the procurement of balancing capacity, forcing the market to procure the given minimum of reserves in the specified geographic area. Relevant for a bidding zones and macro areas in the acquiring domain. For maximum regulation: **A61** = A localization constraint for the procurement of balancing capacity, forcing the market to procure reserves exceeding the maximum regulation outside of the given geographic area. Relevant for bidding zones and macro areas in the acquiring domain. Note: A60 and A61 does not have to be present if any relevant value is not specified.
acquiring_Domain.mRID	M	The identification of the area with the reserve requirement or localization constraint. This can be a bidding zone or a macro area identification.
		A01 - EIC coding scheme
		EIC identification of the market area:

connecting_Domain.mRID	M	**10Y1001A1001A91G** (Nordic Market Area) The EIC identification of the bidding zone where the resources bidding against the reserve requirement are located. This will be set as the Nordic market area in anticipation of a Nordic exchange of FRR balancing capacity. **A01** - EIC coding scheme
quantity_Measurement_Unit.name	M	**MAW** = megawatt
Divisible	M	**A01** = Yes
flowDirection.direction	M	**A01** = Up **A02** = Down
Period		
timeInterval	M	Period covered (in ISO 8601 UTC format). Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ
Resolution	M	**PT60M** – the time resolution.
Point		
Position	M	Sequential value beginning with 1.
quantity.quantity	M	Quantity of: For business type **B75:** Inelastic demand per bidding zone or per macro area For business type **A60:** Minimum reserve regulation per bidding zone or Minimum reserve regulation per macro area (depending on the acquiring domain) For business type **A61:** Maximum reserve regulation per bidding zone or Maximum reserve regulation per macro area (depending on the acquiring domain)

4.7 Bid submission by BSP

The data flow is used by BSPs to submit the bids to the capacity market.

There are two message delivery scenarios. In the first scenario, the BSP sends the message to Nordic MMS by own name. In the second scenario, the agent creates/updates bids on behalf of the other BSP (principal).

Explanatory table:

Sender (sender_MarketParticipant)	subject_MarketParticipant
BSP	BSP
BSP (agent)	BSP (principal)

The ReserveBid_MarketDocument in version 7.1 or 7.4 can be used for bid submission.

For a general description of the ReserveBid_MarketDocument schema in version 7.1, please refer to Ref [5].

For a general description of the ReserveBid_MarketDocument schema in version 7.4, please refer to Ref [6].

Each received document shall always contain data for one day, one BSP and one control area/bidding zone. The scope is therefore defined by combination of the elements:

- reserveBid_Period.timeInterval,
- domain.mRID (= control area/bidding zone),
- subject_MarketParticipant.mRID (= subject BSP),
- auction.mRID of first time series in the document (= market).

The bids within the scope which are already stored in the Nordic MMS are deleted and replaced by new bids in the incoming XML file. In other words, each new XML document always completely replaces all existing bids in the scope (day, control area, subject BSP, market).

Consequently, the submission of the empty XML document would lead to the removal of all existing bids in the scope. However, the empty document does not allow to identify the market in question because the element auction.mRID is on time series level. In that case, the sender must send one "dummy" time series with element status and value A09 = cancelled. The only purpose of this time series is to identify the market. The content of the time series is ignored. The completely empty document will be rejected with the reason code A69 and reason text: *„At least one time series must be present“*.

ReserveBid_MarketDocument		Applicable: 451-7-reservebiddocument.xsd – version 7.1 451-7-reservebiddocument.xsd – version 7.4
mRID	M	Unique identification of the document
revisionNumber	M	Constant value of "**1**"
Type	M	**B40** = Complete Reserve Bid Document
process.processType	M	**A47** = Manual frequency restoration reserve (mFRR) **A51** = Automatic frequency restoration reserve (aFRR)
sender_MarketParticipant.mRID	M	Identification of the party sending the document = valid BSP Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"

sender_MarketParticipant. marketRole.type	M	**A46** = Balancing Service Provider (BSP) **A39** = Data Provider (Agent) Agents sending on behalf of BSPs must use market role A39 when submitting bids. See chapter 6: "Agents and data providers acting on behalf of BSPs"
receiver_MarketParticipant.mRID	M	10V1001C--000284 (Nordic MMS) **A01** = EIC coding scheme
receiver_MarketParticipant. marketRole.type	M	**A34** = Reserve Allocator
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ Used for the general validation that a newer document (i.e. with a more recent createdDateTime) in the same scope has not already been processed.
reserveBid_Period.timeInterval	M	The period covered by the document (in ISO 8601 UTC format) Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ Start and end interval must define an entire CET Day
domain.mRID	M	Code of control area or bidding zone. Must be under the TSO responsibility See list of allowed Bidding zones See list of allowed Control areas Note: If BSP is active in more control areas, it must send more documents – one for each control area.) **A01** – EIC coding scheme

subject_MarketParticipant.mRID	M	Identification of the party responsible for the bid subject_MarketParticipant must be: • the same as the BSP as the sender_MarketParticipant. • or the subject_MarketParticipant must be the principal BSP for the agent BSP which is the sender_MarketParticipant. The relation between principal and agent must be valid and active for the given business day (reserveBid_Period.timeInterval).
		Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
subject_MarketParticipant. marketRole.type	M	**A46** = Balancing Service Provider (BSP)
Bid_TimeSeries		
mRID	M	Unique identification of the time series
auction.mRID	M	Constant value: MFRR_CAPACITY_MARKET AFRR_CAPACITY_MARKET Note: All elements auction.mRID in the document must have the same value. It is not allowed to submit the bids for multiple markets in single file.
businessType	M	**B74** = The time series provides an offer to provide reserves.
acquiring_Domain.mRID	M	**10Y1001A1001A91G** (Nordic Market Area)
		A01 - EIC coding scheme
connecting_Domain.mRID	M	The EIC identification of the bidding zone where the bids are placed. See chapter: 4.3: "Bidding zones". Note: this bidding zone must correspond with domain. If domain is set as control area this bidding zone must below to this control area. If domain is set as bidding zone this biding zone must be the same. This bidding zone must be in the BSP (subject_MarketParticipant.mRID) portfolio.
		A01 - EIC coding scheme

quantity_Measure_Unit.name (v 7.1) **quantity_Measurement_Unit.name (v 7.4)**	M	**MAW** = megawatt
currency_Unit.name	R	**EUR** = euro
price_Measure_Unit.name (v 7.1) **price_Measurement_Unit.name (v 7.4)**	R	**MAW** = megawatt
divisible	M	**A01** = Divisible. Quantity may be reduced stepwise down to the minimum quantity **A02** = Indivisible. No reduction possible on the quantity. Indivisible bids will not be directly allowed for the delivery days from 25.11.2026.
linkedBidsIdentification	D	Used for technical linking of bids by providing a technical linked group identification. Bid time series with common resting time *(resting_ConstraintDuration.duration)* and activation duration limitations *(maximum_ConstraintDuration.duration)* need to be specified with a common technical linked group identification in the linkedBidsIdentification element such that these bids are chosen in market time units where the bids are available for activation. The bid selection will not select such bids in market time units where they can be unavailable due to selections in other market time units of the same or other timeseries with the same linkedBidsIdentification. Not used if the bid is not linked or does not have quality limitations. Quality limitations and thus technical linking cannot be combined with block bids. Allowed for mFRR markets only.

exclusiveBidsIdentification	D	The identification used to associate bids that are to be linked together. If one bid is selected, then all others with the same exclusiveBidsID cannot be selected. Apart from exclusive bid groups within a market and bidding zone, it allows the exclusive cross-linking of bids between markets, to be able to offer a resources flexibility in two markets with simultaneous gate closure without risking double acceptances. Not used if the bid is not part of an exclusive group Note: cannot be used together witch blockBid.
blockBid	O	**A01** – Yes – all quantities within the time interval must be selected **A02** – No Indicator of the block bid. In all MTUs, the block bid must be either accepted (between minimum and offered quantity or fully – depending on the divisible flag) or rejected. I.e. it is not possible to accept the bid in one MTU and reject it in another MTU. If the element is not present, the bid is considered as non-block.
status	O	The information about the status of the bid. The element is optional. If the element is present, it must contain the value **A09** (Cancelled) and is used for being able to cancel all bids for a given delivery date and market. In that case, the content of the time series is ignored. If a time series with such a status is part of the document, it must be the only time series in the document. The time series is present only for the purpose of the identification of the market (which is part of the document scope). As a result of uses of the status element, all bids provided by given BSP (subject_MarketParticipant.mRID) will be deleted from the system. Set of the deleted bids is affected by element domain. • domain.mRID is filled by the control area => bids for given control area are deleted • domain.mRID is filled by the bidding zone => bids for given bidding zone are deleted

registeredResource.mRID	D	Location of the bid. In other contexts this is also called “Resource identifier”, “Location id” and “Stasjonsgruppe id” The element may be either mandatory or disabled for the given market and control area, depending on the local TSO rules. If the element is present, it must contain a valid location information according to the local TSO rules. If the element is not present and the attribute is set as mandatory, it is considered as an error. Please note that for Finland, this identifier is mandatory for mFRR (not for aFRR) and used to indicate the area location of the bid. The coding scheme used is EIC, and the value must be one of these: • "10YFI-0--------3" for North Finland • "10YFI-2--------K" for South Finland • "10YFI-3-------9R" for Central Finland
		Coding scheme: • "NNO": Norwegian national • "NSE": Sweden national
flowDirection.direction	M	**A01** = Up **A02** = Down
marketAgreement.type	O	**A01** – daily
resting_ConstraintDuration. duration	O	The delay to be respected between the end of activation and the start of the next activation. The standard ISO 8601 is used for duration. The allowed form in this context is PTnH, where n is the number of hours. Leading zeros are allowed. The element is optional. If the element is not present, the empty value is written to the corresponding bid attribute. Can be used only for the mFRR market.

maximum_ConstraintDuration.duration	O	The maximum duration that a regulation has to be up once the bid is activated. The standard ISO 8601 is used for duration. The allowed form in this context is PTnH, where n is the number of hours. Leading zeros are allowed. The element is optional. If the element is not present, the empty value is written to the corresponding bid attribute. Can be used only for the mFRR market.
Period		
timeInterval	M	Period covered (in ISO 8601 UTC format) Time interval must be a subset of reserveBid_Period.timeInterval and must not overlap with any other time interval of the same time series. Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ
resolution	M	Constant value PT1H or PT60M The precision of the interval that the different points within the time series cover is *one hour*
Point		
position	M	Position within the time interval. Sequential value beginning with **1**
quantity.quantity	M	Offered quantity
minimum_Quantity.quantity	D (M)	The minimum quantity of energy that can be activated at a given time position. The minimum quantity must be present if bid is marked as divisible (A01) and must not be present when the bid is market as indivisible (A02). Since delivery day 25.11.2026, this element is mandatory because completely indivisible bids with variable volumes across MTUs will no longer be supported. Must be specified as the same value across all points across all periods of the bid.
price.amount	M	The price of the product offered Must be specified as the same value across all points across all periods of the bid.

4.8 Publishing bids results by Nordic MMS

The data flow is used for distribution of detailed results (accepted and not accepted bids) towards BSPs after auction clearing.

There are two message delivery scenarios. In the baseline scenario, the BSP receives the message as expected. In the alternative scenario, an agent acting on behalf of the BSP receives the message instead of the affected BSP (principal).

Explanatory table:

Receiver	Party in time series	Description
BSP	BSP	Results are distributed directly to BSP which is the input of the dataflow.
BSP (agent)	BSP (principal)	Results are distributed to the agents – the data of their principals are sent. BSP (agent) always receives one document containing all bids in all control areas of their principals).

The document is sent to:

- BSPs who have inserted at least one bid.
- Agent on behalf of the principal BSP if at least one bid of its principal exists.

List of possible documents:

The publishing of detailed results using the ReserveAllocationResult_MarketDocument in version 6.4.

For a general description of the ReserveAllocationResult_MarketDocument schema in version 6.4, please refer to Ref [8].

ReserveAllocationResult_ MarketDocument		iec62325-451-7-reserveallocationresult.xsd – version 6.4
mRID	M	Unique identification of the document
revisionNumber	M	Constant value of "**1**"
type	M	**A38** = Reserve allocation result document
process.processType	M	**A47** = Manual frequency restoration reserve (mFRR) **A51** = Automatic frequency restoration reserve (aFRR)
sender_MarketParticipant.mRID	M	10V1001C—000284 (Nordic MMS) **A01** = EIC coding scheme
sender _MarketParticipant. marketRole.type	M	**A34** = Reserve Allocator
receiver_MarketParticipant.mRID	M	Identification of the party receiving the document Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"

receiver_MarketParticipant. marketRole.type	M	**A46** = Balancing Service Provider (BSP) **A39** = Data Provider, see chapter 6: "Agents and data providers acting on behalf of BSPs"
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
reserveBid_Period.timeInterval	M	Period covered (in ISO 8601 UTC format) YYYY-MM-DDTHH:MMZ
domain.mRID	M	Code of control area. Each document covers just one control area. If BSP is active in more control areas, it receives more documents – one for each control area.
		A01 - EIC coding scheme
TimeSeries **Each time series represent one bid. However, each bid may be presented by:** • **one time series for accepted quantities for balancing purposes (standard output of the market clearing) – this time series is always present for the bid,** • **one for accepted quantities due to the bottleneck – present only if the bid is at least partially accepted due to the bottleneck in the transmission network,** • **one for rejected quantities due to the bottleneck – present only if the bid is at least partially rejected due to the bottleneck in the transmission network,** **Each time series (standard and bottleneck) can contain the different time interval.**		
mRID	M	An identification that uniquely identified the time series within the document. Generated automatically by Nordic MMS. This is not an identification of the bid.
bid_Original_MarketDocument. mRID	M	Constant value of "**NA**"
bid_Original_MarketDocument. revisionNumber	M	Constant value of "**1**"
bid_Original_MarketDocument. bid_BidTimeSeries.mRID	M	The bid ID The identification of the time series that was used in the original tender – the reference to the bid
bid_Original_MarketDocument. tendering_ **MarketParticipant.mRID**	M	The ID of the tendering party
		Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"

auction.mRID	M	Possible values: MFRR_CAPACITY_MARKET AFRR_CAPACITY_MARKET
businessType	M	**B95** = Procured capacity (used for Auction contracts)
acquiring_Domain.mRID	M	**10Y1001A1001A91G** (Nordic Market Area) **A01** = EIC coding scheme
connecting_Domain.mRID	M	The EIC identification of the bidding zone where the bids were placed. See chapter: 4.3: "Bidding zones". **A01** = EIC coding scheme
marketAgreement.type	M	**A01** = daily
marketAgreement.mRID	M	Constant value of "**NA**"
quantity_Measurement_Unit.name	M	**MAW** = megawatt
currency_Unit.name	R	**EUR** = Euro
price_Measurement_Unit.name	M	**MAW** = megawatt
registeredResource.mRID	O	The identification of a location associated with a bid. If location is not used, the element is omitted.
flowDirection.direction	M	**A01** = Up **A02** = Down
resting_ConstraintDuration. duration	O	The delay to be respected between the end of activation and the start of the next activation. The standard ISO 8601 is used. The number of hours stored in the time series is converted to the format PTnH, where n is the number of hours. If the value of the time series is empty, the element is omitted.
maximum_ConstraintDuration. duration	O	The maximum duration that a regulation has to be up once the bid is activated. The standard ISO 8601 is used. The number of hours stored in the time series is converted to the format PTnH, where n is the number of hours. If the value of the time series is empty, the element is omitted.

Reason (within Time Series) **Two reason codes are specified.**		
code	M	The first code informs whether the presented values describe bid from the perspective of standard balancing process or acceptance due to bottleneck. **B49** = bid is described by the perspective of the standard balancing process **B42** = the time series describes the bid in a bottleneck perspective (accepted or rejected due to the bottleneck).
code	M	The second code informs whether the bid has been accepted or rejected. **A73** = bid has been fully accepted (in all hours of the day) **A72** = bid has been partially accepted (at least in one hour). If the bid is partially accepted for balancing purposes and partially due to the bottleneck (and consequently, the bid is present twice in the document), both time series are marked as partially accepted. **B09** = bid has been rejected by the clearing algorithm **B16** = bid has been rejected by TSO (in combination with reason code B49 above) or due to the bottleneck (in combination with reason code B42 above).
text	D	A reason for rejection described in flat text. A reason for rejection can be specified by the TSO and will be included here. Only may be present if reason code = B16

<table>
<tr><th colspan="3">Period</th></tr>
<tr><td>timeInterval</td><td>M</td><td>Period covered (in ISO 8601 UTC format)

YYYY-MM-DDTHH:MMZ

Each time interval represents one continuous interval of hours where accepted quantity is available.

<table>
<tr><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>Etc.</th></tr>
<tr><td>V</td><td>V</td><td>V</td><td>V</td><td></td><td></td><td>V</td><td>V</td><td></td><td>Etc.</td></tr>
</table>
"V" represents any filled-in accepted quantity.

In this case, the following intervals are built:

• 1–4 (i.e. 00:00–4:00)
• 7–8 (i.e. 6:00–8:00)

For the time series representing the acceptance for the balancing purposes (which is present always in the document), the accepted quantity is available for each hour for which there is a bid quantity. The accepted quantity is always filled with at least zero (in case of non-accepted quantity).

For the time series representing the acceptance due to the bottleneck (which is present optionally), the accepted quantity is present only for hours where the bid is at least partially accepted due to the bottleneck.</td></tr>
<tr><td>resolution</td><td>M</td><td>PT60M – the precision of the interval that the different points within the time series cover is one hour</td></tr>
<tr><th colspan="3">Point</th></tr>
<tr><td>position</td><td>M</td><td>Position within the time interval. Sequential value beginning with 1</td></tr>
<tr><td>quantity</td><td>M</td><td>The accepted quantity of the bid in the given point.

For combination of reason codes B42 and B16 (rejection due to the bottleneck), the element contains the quantity rejected due to the bottleneck.</td></tr>
<tr><td>price.amount</td><td>D</td><td>The accepted price of the contract identified for a point

The price is present if accepted quantity > 0. The price is not present for rejected quantity.

The accepted price can be different from the original offered price and can be different hour by hour.

The quantity accepted for balancing purposes is charged by the clearing price. The quantity accepted due to the bottleneck is charged by the bid price.</td></tr>
</table>

secondaryQuantity	R	**The offered quantity** identified for a point
bid_Price.amount	R	**The offered price** identified for a point

4.9 Publishing total market results by Nordic MMS

The data flow is used to distribute overall auction results (area volumes and prices) towards BSPs. The document is sent to all BSPs who submitted at least one bid (in any bidding zone). The document contains the data of all bidding zones in the given market.

The publishing of market results using the Balancing_MarketDocument in version 4.5.

For a general description of the Balancing_MarketDocument schema in version 4.5, please refer to Ref [7].

Balancing_MarketDocument		**iec62325-451-6-balancing.xsd – version 4.5**
mRID	M	Unique identification of the document
revisionNumber	M	Constant value of "**1**"
type	M	**B34** = Market result document
process.processType	M	**A47** = Manual frequency restoration reserve (mFRR) **A51** = Automatic frequency restoration reserve (aFRR)
sender_MarketParticipant.mRID	M	10V1001C—000284 (Nordic MMS) **A01** = EIC coding scheme
sender _MarketParticipant.marketRole.type	M	**A34** = Reserve Allocator
receiver_MarketParticipant.mRID	M	Identification of the BSP receiving the document Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	M	**A46** = Balancing Service Provider (BSP) **A39** = Data Provider, see chapter 6: "Agents and data providers acting on behalf of BSPs"
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
area_Domain.mRID	M	10Y1001A1001A91G (Nordic Market Area) **A01** – EIC coding scheme

period.timeInterval	M	The period covered by the document (in ISO 8601 UTC format) YYYY-MM-DDTHH:MMZ
TimeSeries		
mRID	M	An identification that uniquely identifies the time series
businessType	M	**C17** = Market price and volume
acquiring_Domain.mRID	M	**10Y1001A1001A91G** (Nordic Market Area) **A01** - EIC coding scheme
connecting_Domain.mRID	M	The EIC identification of the bidding zone. See chapter: 4.3: "Bidding zones". **A01** - EIC coding scheme
type_MarketAgreement.type	M	**A01** = daily
flowDirection.direction	M	**A01** = Up **A02** = Down
currency_Unit.name	R	**EUR** = Euro
quantity_Measurement_Unit.name	R	**MAW** = megawatt
price_Measurement_Unit.name	R	**MAW** = megawatt
auction.mRID	R	Possible values: MFRR_CAPACITY_MARKET AFRR_CAPACITY_MARKET
Period		
timeInterval	M	Period covered (in ISO 8601 UTC format) YYYY-MM-DDTHH:MMZ
Resolution	M	**PT60M** = the precision of the interval that the different points within the time series cover is one hour
Point		
Position	M	Position within the time interval. Sequential value beginning with **1**
Quantity	M	Total volume procured
procurement_Price.amount	M	Market price

4.10 Acknowledgement document returned by Nordic MMS

Acknowledgement_MarketDocument		iec62325-451-1-acknowledgement.xsd – v 8.0
mRID	M	Unique identification of the document
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
sender_MarketParticipant.mRID	M	10V1001C--000284 **A01** = EIC coding scheme
sender _MarketParticipant.marketRole.type	M	**A34** = Reserve Allocator
receiver_MarketParticipant.mRID	M	Identification of the BSP receiving the ACK Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	M	**A46** = Balancing Service Provider (BSP) **A39** = Data Provider, see chapter 6: "Agents and data providers acting on behalf of BSPs"
received_MarketDocument.mRID	M	Information extracted from the bid document
received_MarketDocument. revisionNumber	M	Information extracted from the bid document
received_MarketDocument. createdDateTime	M	Information extracted from the bid document
Rejected_TimeSeries		
mRID	D	Identification of the rejected time series
Reason	D	One or more of the reasons described below
code	M	**A01** = Message fully accepted **A02** = Message fully rejected **A05** = Sender without valid contract. Used if the BSP is not eligible for the given product in the current bid area **A57** = Deadline limit exceeded/Gate not open **A59** = Not compliant to local market rules Other reason codes may also be used.

text	M	A specific reason for the rejection can be specified and included here. E.g.: «Time series ID»: The exclusive group must contain at least two bids

5 ECP

ECP is the integration channel that is supported for communication with the Nordic MMS market platform. It is used for all data exchanges between Nordic MMS and BSPs.

5.1 How it works

ECP delivers messages from the sender to a recipient within single ECP network. Messages transported through the ECP network can be any text or binary data. IEC CIM is used as data exchange format. Alongside with the message, ECP also transfers message metadata. These are (among others) information about sender and recipient. The former is used by platform to authenticate the message sender. ECP consists of three main components: *endpoint*, *component directory* and *broker*. See figure below:

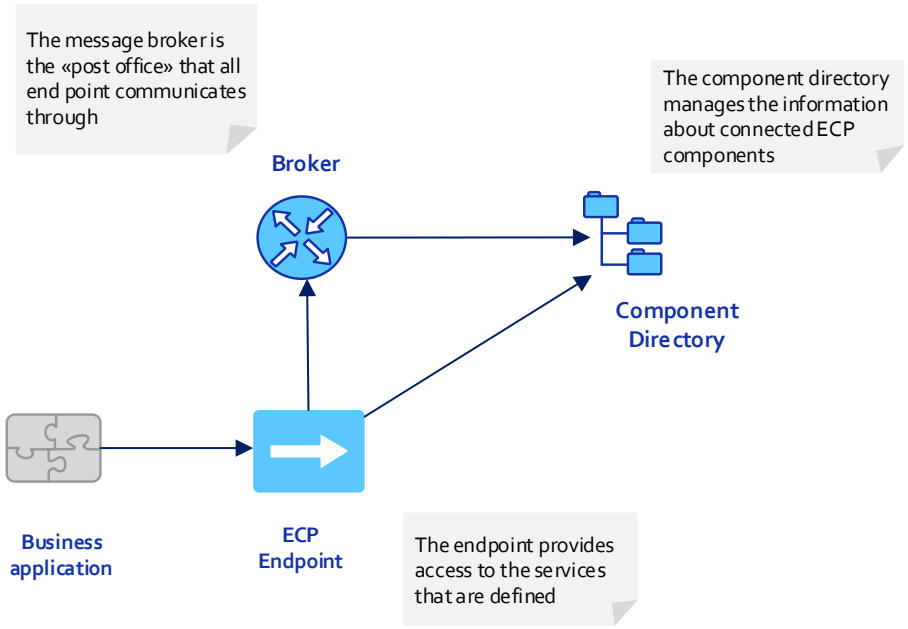
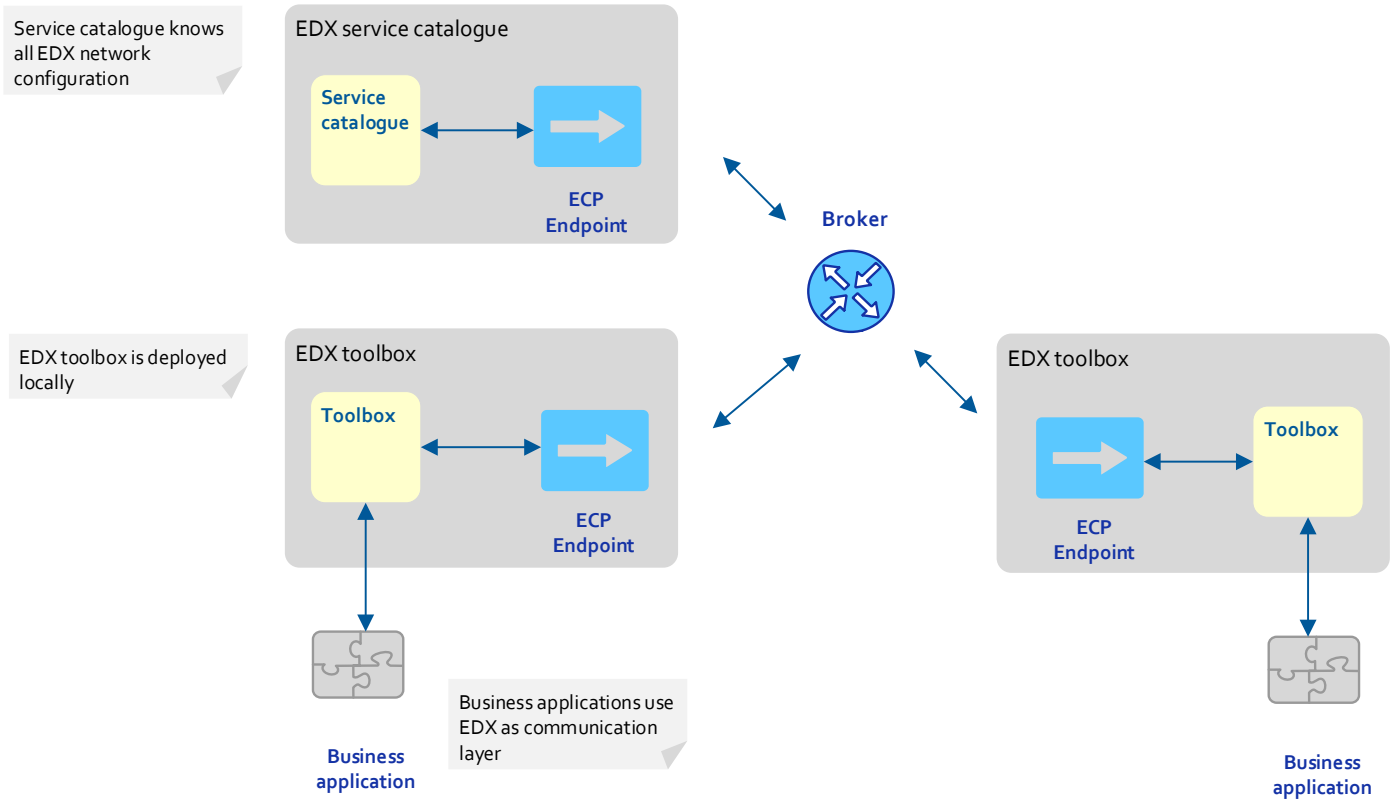


Figure 2 ECP Main Components

5.2 EDX

EDX is an extension to ECP, and is used to define the network configuration, and introduces the concept of services, service providers and consumers. The two central parts of the EDX network is the service catalogue and the toolbox. An EDX network consists of multiple toolboxes and a single service catalogue.

These components communicates via ECP and is responsible for distribution of the network configuration. Messaging occurs directly between the toolboxes. Se figure below for EDX network overview:



5.3 How to connect

Please refer to the installation package and documentation provided by your local TSO.

5.4 ECP/EDX for capacity market

To see list of the ECP message types used for Nordic MMS, please visit chapter 2.4

The addressing convention used for communicating in the ECP network is *SERVICE-service.code*.

Note: Nordic MMS does not validate ECP service name but it is recommended to use values described in the following table.

Market	Market code	EDX service	ECP Service name
mFRR capacity market	MFRR_CAPACITY_MARKET	NO-MFRRCAP	SERVICE-NO-MFRRCAP
aFRR capacity market	AFRR_CAPACITY_MARKET	AFRRCAP	SERVICE-AFRRCAP

6 Agents and data providers acting on behalf of BSPs

Nordic MMS supports authorizing third parties (agents) and their users to bid and receive market results on behalf of one or more principal BSPs (the party delegating the right to bid and receive market results).

The authorized parties can, but do not have to have own portfolios in Nordic MMS.

The authorization scheme allows the agent's ECP endpoint to be used to send bids and receive market results for one or more subject party (principal BSP).

For this purpose, the endpoint must belong to and identify the agent acting as a data provider on behalf of the principal BSPs. The agent's party code has to be set as the sender_MarketParticipant.mRID in the message and its sender_MarketParticipant.marketRole.type needs to be set to A39 – data provider.

In order to authorize an agent, the principal BSP needs to request an authorization for the agent party from its responsible TSO. The BSP needs to provide both the agent's party code and ECP endpoint. An agent can only be authorized for all or none of the principal BSPs bidding zones.

Once the association between the agent and the principal BSP is defined, the agent's users will have access to the bids and market results of the principal BSP. The agent's ECP endpoint will be allowed to submit bids and receive market results on behalf of the principal BSP.